

Chapter 1 Overview

	1.1 Systems of Linear Equations
Learning Outcomes	- What is a linear equation and a linear system? - What is a general solution of a linear equation/system? - What is the geometrical interpretation of a linear equation/system and its solutions? - How many solutions can a linear system have? - What is a homogeneous system? - What are trivial and non-trivial solutions of a homogeneous system?
Key Points	- A linear system has either no solution, exactly one solution, or infinitely many solutions. - A homogeneous system has either only one (trivial) solution, or infinitely many solutions (trivial and non-trivial solutions) - A homogeneous system of linear equations with more variables than equations has infinitely many solutions.
	1.2 Solving System of Linear Equations
Learning Outcomes	- What are the three elementary row operations (ERO) and their notations? - How to perform ERO? - What is a row-echelon form (REF) of an augmented matrix? - What are leading entries and pivot columns of a REF? - How to use REF to get solutions of linear system? - How to tell the number of solutions from a REF? - How are the general solutions of a non-homogeneous system and its associated homogeneous system related?
Key Points	- The solutions of a LS can be obtained from its REF - When every column of a REF is a pivot column (except the last column), the system has exactly one solution. - When there is at least one non-pivot column in a REF (other than the last column), the system has infinitely many solutions. - When the very last column of a REF is a pivot column, the system has no solution. - The general solution of the non-homogeneous system is the sum of the general solution of the associated homogeneous system and a particular solution of (A).
	1.3 Gaussian Elimination
Learning Outcomes	- What are Gaussian elimination (GE) and Gauss-Jordan elimination (GJE)? - What is a reduced row-echelon form (RREF)? - How to use GE to reduce an augmented matrix to a REF? - How to use GJE to reduce an augmented matrix to a RREF?
Key Points	- RREF is a special type of REF with additional conditions on each pivot column - The process of GE is completed when a REF is obtained. - The process of GJE is completed only when a RREF is obtained. - The solutions of a linear system obtained using either GE or GJE are the same.

	1.4 Matrices
Learning Outcomes	- What are the size and entries of a matrix? - What are diagonal, identity, symmetric, triangular matrices? - How to perform matrix addition, matrix multiplication, scalar multiplication, powers and transpose? - What are some properties of matrix operations?
Key Points	- Matrix multiplication: $\mathbf{AB} \neq \mathbf{BA}$ (in general) - Matrix multiplication: $\mathbf{AB} = \mathbf{0}$ does not imply $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$ - Matrix multiplication satisfies associative and distributive properties - Zero matrix and identity matrix behave like numbers "0" and "1" respectively in matrix multiplication. - Matrix transpose: $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ $\mathbf{A}$ is a symmetric matrix if and only if $\mathbf{A} = \mathbf{A}^T$
	1.5 Matrix inverses I
Learning Outcomes	- How to express linear system in matrix equation form? - What is a non-singular (invertible) matrix? - What is the inverse of a matrix? - How to find inverse of a 2 x 2 matrix? - How to solve linear system in matrix equation form?
Key Points	- Linear system and its solutions can be expressed in matrix equation form - Linear system can be denoted as $\mathbf{Ax} = \mathbf{b}$ with coefficient matrix $\mathbf{A}$, variable matrix $\mathbf{x}$ and constant matrix $\mathbf{b}$. - We can solve a linear system in matrix equation form $\mathbf{Ax} = \mathbf{b}$ if the coefficient matrix $\mathbf{A}$ is a non-singular square matrix, and the solution is given by $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ - Not every square matrix has an inverse. - A 2 x 2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is non-singular if $ad - bc \neq 0$.
	1.6 Matrix inverses II
Learning Outcomes	- How to find the inverse of an n x n non-singular matrix? - How to determine whether an n x n matrix is singular or not? - What is the relation between the singularity of a square matrix and its REF? - What are the effects of some matrix operations on non-singular matrices? - What is the relation between the singularity of a matrix $\mathbf{A}$ and the solutions of the linear system $\mathbf{Ax} = \mathbf{b}$?
Key Points	- Inverse of a non-singular matrix can be found using GJE. - If a sequence of elementary row operations reduce $\mathbf{A}$ to the identity matrix $\mathbf{I}$, the same sequence will also reduce $\mathbf{I}$ to $\mathbf{A}^{-1}$. - If the RREF of $\mathbf{A}$ is an identity matrix $\mathbf{I}$, then $\mathbf{A}$ is non-singular. - If any REF of $\mathbf{A}$ has a zero row, then $\mathbf{A}$ is singular. - If $\mathbf{A}$ is non-singular, then $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$ - If $\mathbf{A}, \mathbf{B}$ are non-singular, then $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ - If $\mathbf{A}$ is non-singular, the linear system $\mathbf{Ax} = \mathbf{b}$ has exactly one solution - If $\mathbf{A}$ is singular, the linear system $\mathbf{Ax} = \mathbf{b}$ has either no solution or infinitely many solutions

	1.7 Determinants of Matrices
Learning Outcomes	- What is the determinant of a matrix and how to compute it? - How do matrix operations affect determinants? - How do elementary row operations affect determinants? - What is the relation between singularity and determinant? - What are the determinants of some special matrices?
Key Points	- Determinant of an $n \times n$ matrix can be computed inductively by expanding it as the alternating sum of its $(n-1) \times (n-1)$ submatrices. - Determinant of a triangular (diagonal) matrix is the product of its diagonal entries $\det(\mathbf{A}) = \det(\mathbf{A}^T)$ - If $\mathbf{A}$ has two identical rows (columns), then $\det(\mathbf{A}) = 0$ - Interchanging two rows/columns changes determinant by a negative sign - Adding a multiple of a row (column) to another will not change determinant. $\mathbf{A}$ is non-singular if and only if $\det(\mathbf{A}) \neq 0$ - If $\mathbf{A}$ is $n \times n$ and c is a scalar, then $\det(c\mathbf{A}) = c^n \det(\mathbf{A})$ $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$
	1.8 Ranks of Matrices
Learning Outcomes	- What is the rank of a matrix? - What are the possible ranks of a $m \times n$ matrix? - What is a full rank matrix? - What is the relation between rank and determinant of a matrix?
Key Points	- If $\mathbf{R}$ is a REF of $\mathbf{A}$, then $\text{rank}(\mathbf{A}) = \text{number of non-zero rows in } \mathbf{R} = \text{the number of leading entries in } \mathbf{R} = \text{the number of pivot columns in } \mathbf{R}$ - Largest possible rank of an $m \times n$ matrix is $\min\{m, n\}$ - An $n \times n$ square matrix $\mathbf{A}$ is full rank $\Leftrightarrow \text{rank}(\mathbf{A}) = n \Leftrightarrow$ the RREF of $\mathbf{A}$ is the identity matrix $\Leftrightarrow \mathbf{A}$ is non-singular $\Leftrightarrow \det(\mathbf{A}) \neq 0$